Hydrofarms

PRODUCTION AND NURSERY TABLES

Fabrication drawings, steel sections, connection details, cutting list, galvanizing requirements and prototype test plan.

OVERALL SIZE	PRODUCTION	NURSERY
16,000 x 2,000 mm	10 NFT channels	17 NFT channels

UNIFIED CHASSIS MASS incl. galvanizing allowance

285.1 kg / table

PROTOTYPE FABRICATION ISSUE - NO MASS PRODUCTION BEFORE TEST AND APPROVAL

1. Design Basis and Table Comparison

Both table types use one steel chassis to reduce cost and spare parts. Only the NFT profile, channel count and clamp set change. Common transverse supports at 1,000 mm centres eliminate a separate longitudinal rail below every channel.

Item	Production	Nursery
Length x width	16,000 x 2,000 mm	16,000 x 2,000 mm
NFT channels	10 x 150 x 57 x 2.2 mm	17 x 100 x 50 x 2.0 mm
Channel pitch	200 mm	115 mm
Plant holes	80 / channel; 800 / table	159 / channel; 2,703 / table
Support lines	17 @ 1,000 mm	17 @ 1,000 mm
Leg portals	9 @ 2,000 mm	9 @ 2,000 mm
Longitudinal fall	1% (160 mm total)	1% (160 mm total)
Project quantity	78 tables	6 tables

FEED-END LEVEL

900 mm

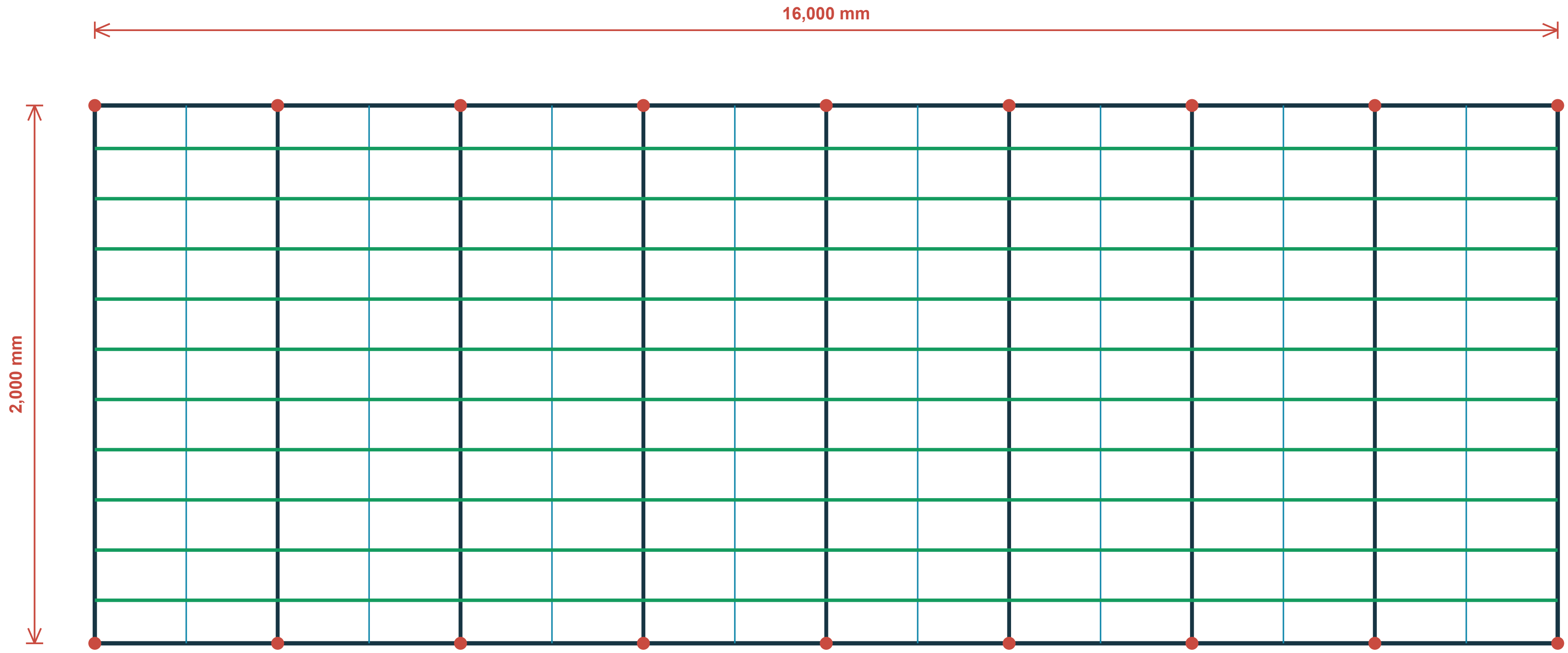
DRAIN-END LEVEL

740 mm

DISTRIBUTED TEST LOAD

1,200 kg / 24 h

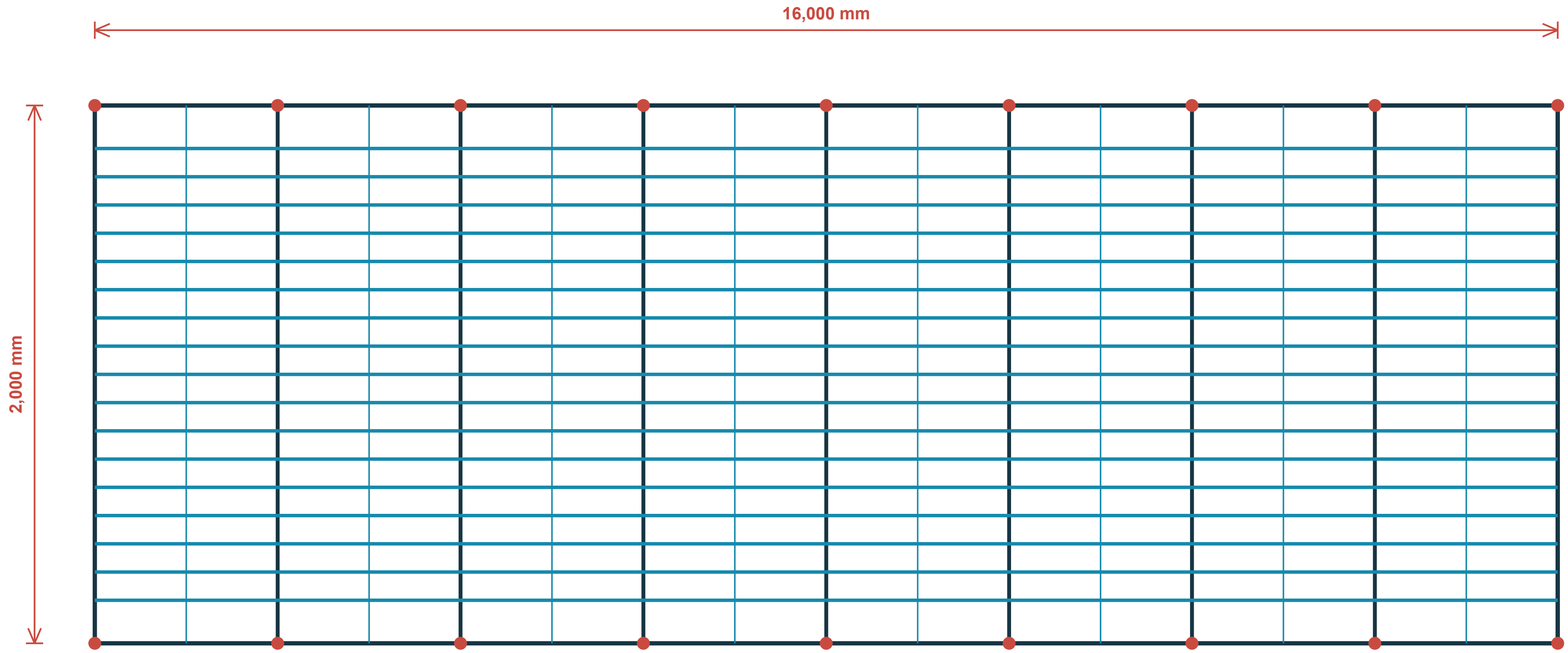
2. PRODUCTION TABLE PLAN



17 TRANSVERSE SUPPORT LINES @ 1,000 mm - LEGS @ 2,000 mm

CHANNEL COUNT	CENTRE PITCH	NFT PROFILE
10	200 mm	150 x 57 x 2.2 mm

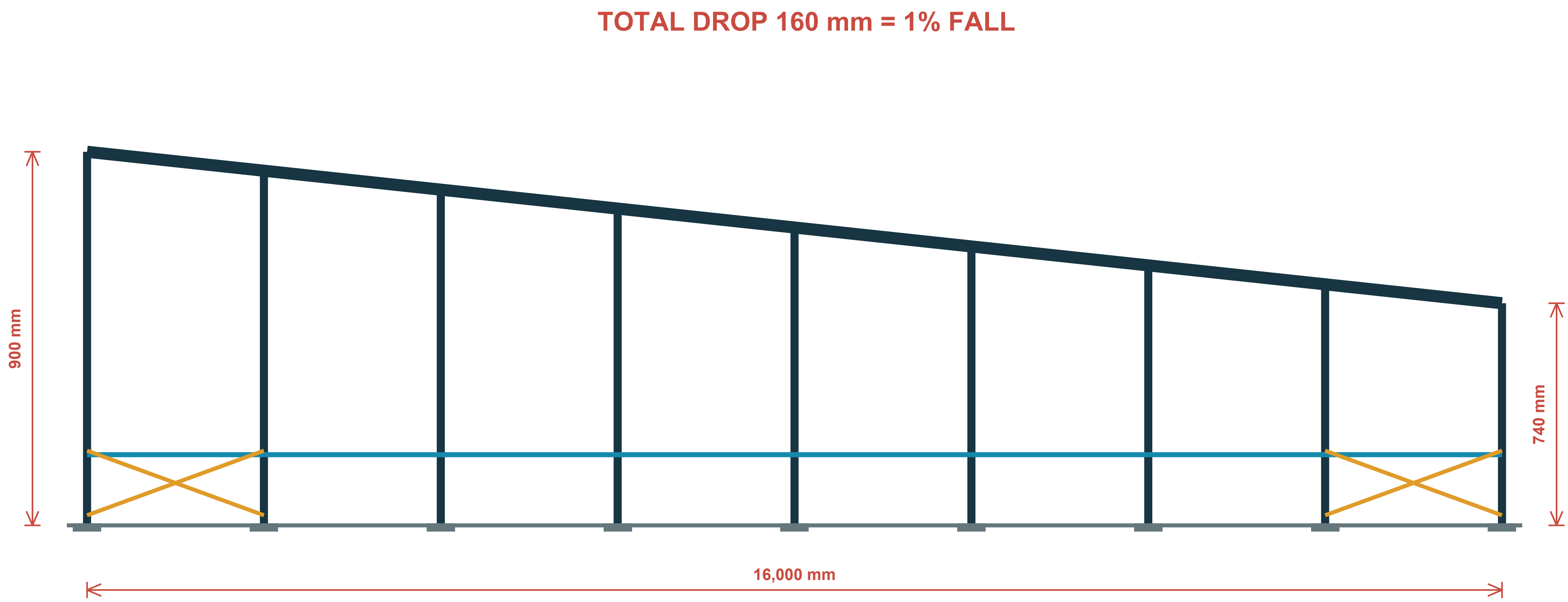
2. NURSERY TABLE PLAN



17 TRANSVERSE SUPPORT LINES @ 1,000 mm - LEGS @ 2,000 mm

CHANNEL COUNT	CENTRE PITCH	NFT PROFILE
17	115 mm	100 x 50 x 2.0 mm

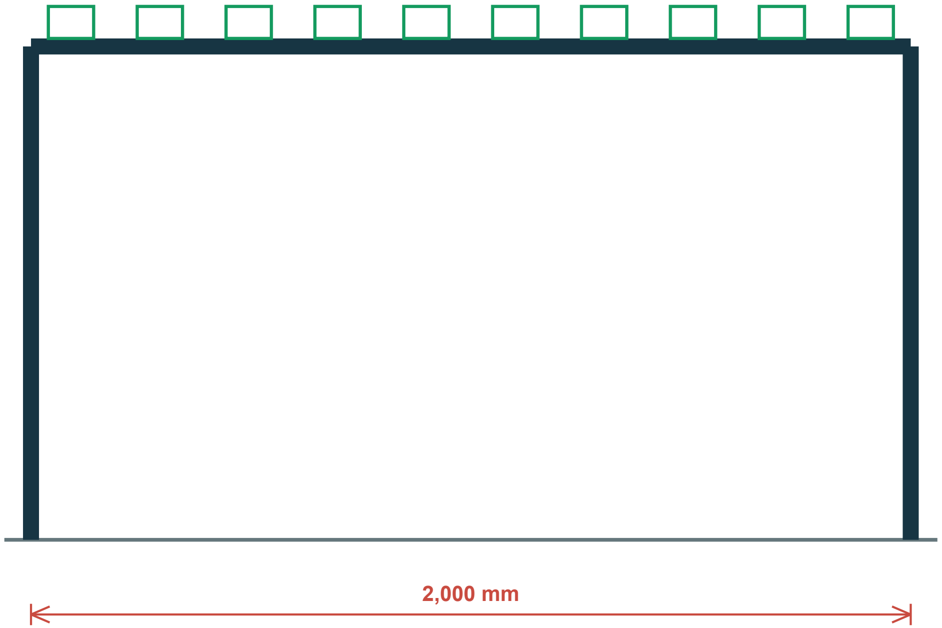
3. Long Elevation and Fall Setting



Cut legs to the scheduled levels or use approved galvanized levelling inserts. Loose timber or plastic shims are prohibited. Provide X-bracing in the first and last bay on both sides.

4. End Elevations and NFT Channel Set-out

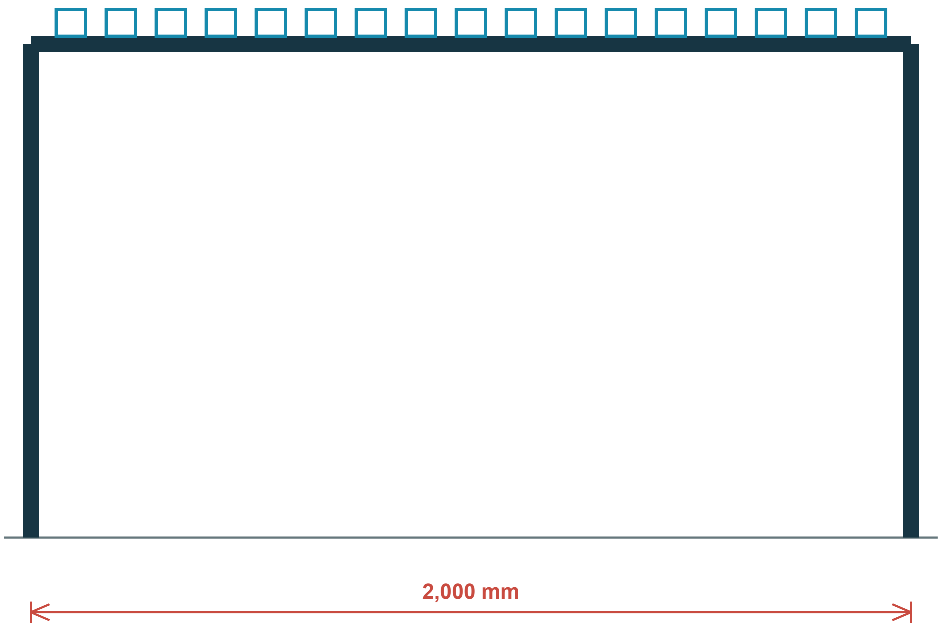
PRODUCTION TABLE



10 @ 200 mm c/c

NFT 150 x 57 x 2.2 mm

NURSERY TABLE



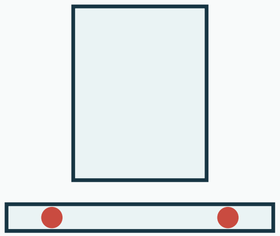
17 @ 115 mm c/c

NFT 100 x 50 x 2.0 mm

CONTINUOUS EPDM ON EACH SUPPORT + CLAMP AT EVERY CROSSING; DO NOT DRILL CHANNEL BOTTOM

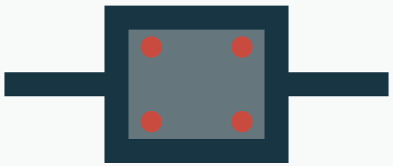
5. Connections and Fixing Details

D1 **FOOT PLATE**



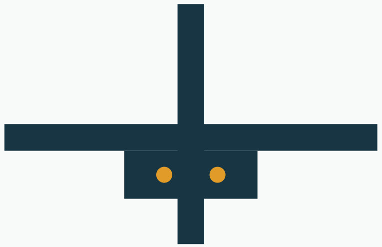
SHS 40x40x2 / PL 120x120x5

D2 **BEAM SPLICE**



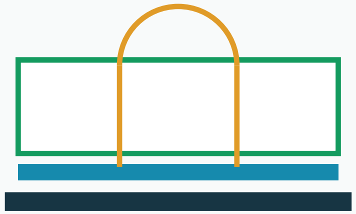
2 x PL 80x200x5 + 4 x M10

D3 **CROSS-BAR CLEAT**



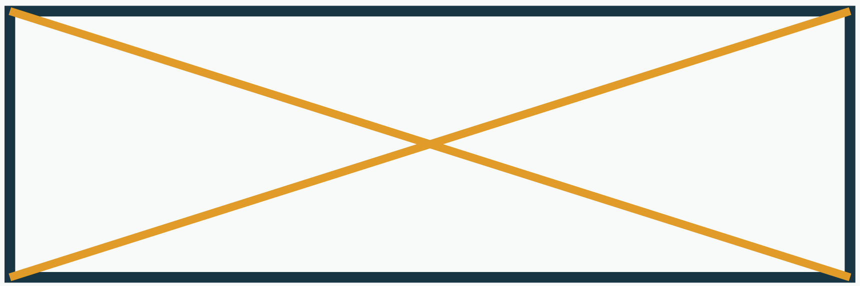
L40x40x4x100 + 2 x M10

D4 **NFT CLAMP**



EPDM + CLAMP; NO BOTTOM HOLE

D5 **X BRACING**



Flat 30x3 / M10 end bolts

D6 **GALVANIZING VENTS**

Vent and drain every sealed hollow section before galvanizing. Complete cutting, drilling and welding in the workshop before hot dipping.

6. Per-Chassis Cutting List and Mass

Code	Description	Section	Cuts	Length m	kg
S01	Side longitudinal beams	SHS 40x40x2	8 x 4,000	32.00	76.4
S02	Graded legs	SHS 40x40x2	18 pcs	14.76	35.2
S03	Transverse supports	SHS 40x40x2	17 x 2,000	34.00	81.1
S04	Lower side ties	SHS 25x25x1.5	8 x 4,000	32.00	35.4
S05	Diagonal bracing	Flat 30x3	Allowance	22.60	16.0
S06	Foot plates	PL 120x120x5	18 pcs	-	10.2
S07	Side-beam splice plates	PL 80x200x5	12 pcs	-	7.5
S08	Bar and tie cleats	L40x40x4x100	50 pcs	5.00	11.9
S09	Bolts, nuts, washers	M10 Gr 8.8 HDG	Allowance	-	3.0

BARE STEEL

276.8 kg

INCL. 3% GALV. ALLOWANCE

285.1 kg

PROCUREMENT LIMIT

295 kg max

84 CHASSIS TOTAL: APPROX. SHIPPING MASS 23.94 t

7. Fabrication, Galvanizing and Assembly

Item	Requirement
Material	S235JR minimum structural steel or approved equivalent; actual wall thickness not below schedule.
Welding	Weld to AWS D1.1 or equivalent; typical 4 mm fillet; remove slag and burrs.
Galvanizing	Hot-dip after fabrication to ISO 1461 or ASTM A123, 75 micron minimum average.
Bolting	M10 grade 8.8 HDG with flat and spring washers; do not crush hollow sections.
Tolerances	Length +/-5 mm; width +/-3 mm; support level +/-2 mm; final fall +/-0.1%.
Assembly	Assemble four 4 m modules; no site welding after galvanizing; anchor only at engineer-marked locations.
Isolation	EPDM between NFT and steel; closed ends; no sharp contact and no channel-bottom drilling.

ASSEMBLY: MAIN PORTALS -> BEAMS & TIES -> INTERMEDIATE BARS -> X-BRACING -> FALL SETTING -> EPDM & CLAMPS -> NFT CHANNELS

8. Prototype Test and Handover

1		2		3		4		5
DIMENSIONS / FALL	->	ACTUAL CHANNELS	->	1,200 kg LOAD	->	24 h HOLD	->	MEASURE / APPROVE

Acceptance item	Value / signature
Maximum loaded deflection	6.5 mm or L/300, whichever is stricter
Residual deflection	2.0 mm maximum
Connections	No weld cracks, loosening or local buckling
Water run	Complete drainage with no ponding or unacceptable sag
Prototype mass after galvanizing	________ kg (295 kg maximum)
Hydrofarms approval	________________________
Consultant approval	________________________

NOTICE: Prototype package, not a substitute for stamped structural approval. Do not fabricate the remaining 83 tables before test and sign-off.